pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Decision Trees

WEEK 2 · DAY 4

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

A model that asks questions

Ask

a question of one column

Split

the accounts in two

Stop

the hard part: today's lesson

Allowed names

model probability of default in this file

flagged at threshold t

the model's score

Never called

a credit decision or an approval

a lending policy or a lending rule

a risk grade for a real person

Real UCI data: 30,000 accounts, Taiwan, 2005. No person column is a feature. Test rows stay closed.

A tree as questions

Word	Meaning
Node	A group of training accounts
Root	The first node: every account
Split	Is this value at most t? Yes left
Leaf	A node with no question
Depth	Questions from root to deepest leaf

```
|--- PAY_0 <= 1.50
|   |--- weights: [13455.00, 2660.00] class: 0
|--- PAY_0 >  1.50
|   |--- weights: [563.00, 1322.00] class: 1
```

18,000 training accounts. Left: codes -2, -1, 0, 1 (two undocumented). Right: a delay of two months or more.

PAY_0 <= 1.5: how this file splits, in its training rows. Not a rule for lending.

Gini impurity, and a split by hand

```
Gini = 1 - p^2 - (1 - p)^2     p = share of defaults;  0 pure, 0.5 half and half
```

Group	Accounts	Defaults	Share	Gini
all training rows	18,000	3,982	0.2212	0.3446
left: PAY_0 <= 1.5	16,115	2,660	0.1651	0.2756
right: PAY_0 > 1.5	1,885	1,322	0.7013	0.4189

0.2906

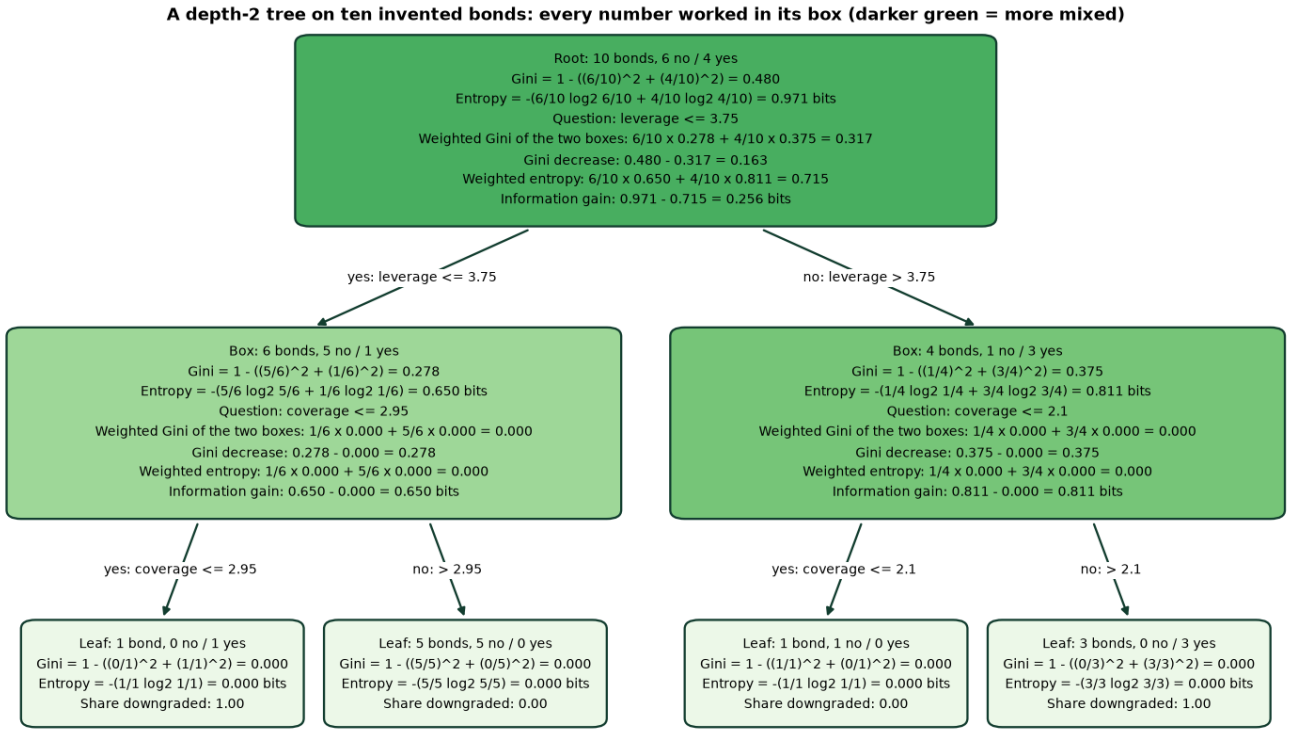
weighted Gini of the two sides

0.0539

decrease: the largest of any question

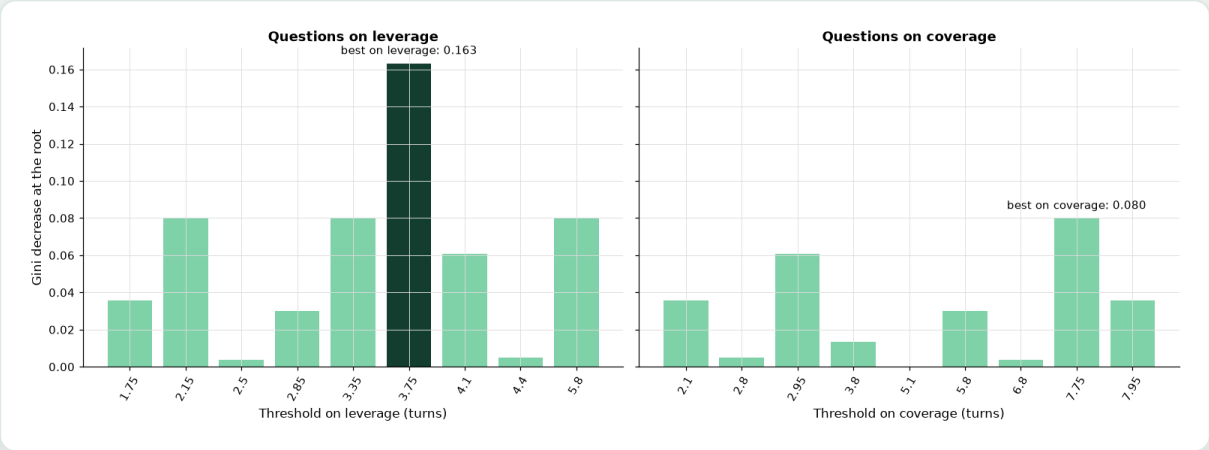
Four counts by COUNTIFS. Excel does it too.

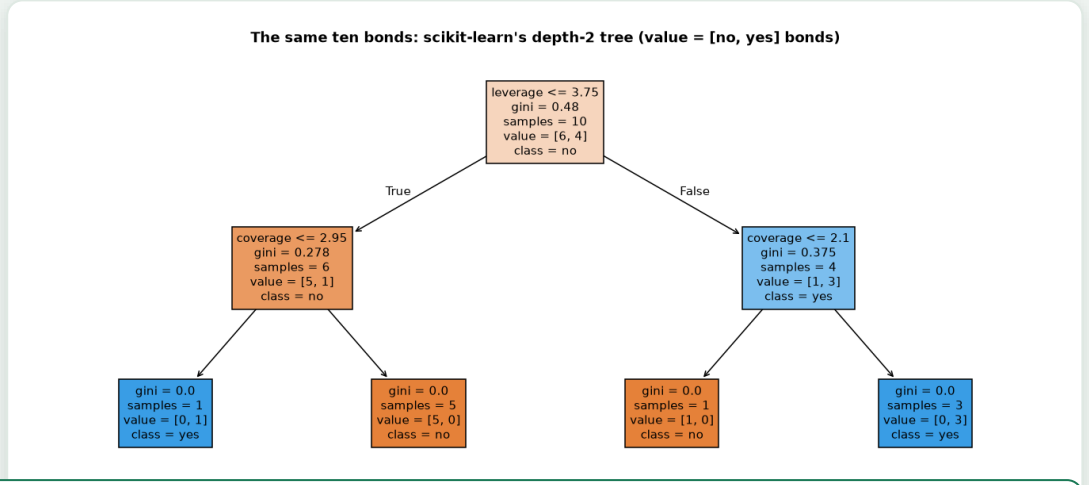
Gini in boxes: ten invented bonds



Root: Gini 0.480, question leverage <= 3.75, weighted Gini 0.317, decrease 0.163. Pure boxes score 0.

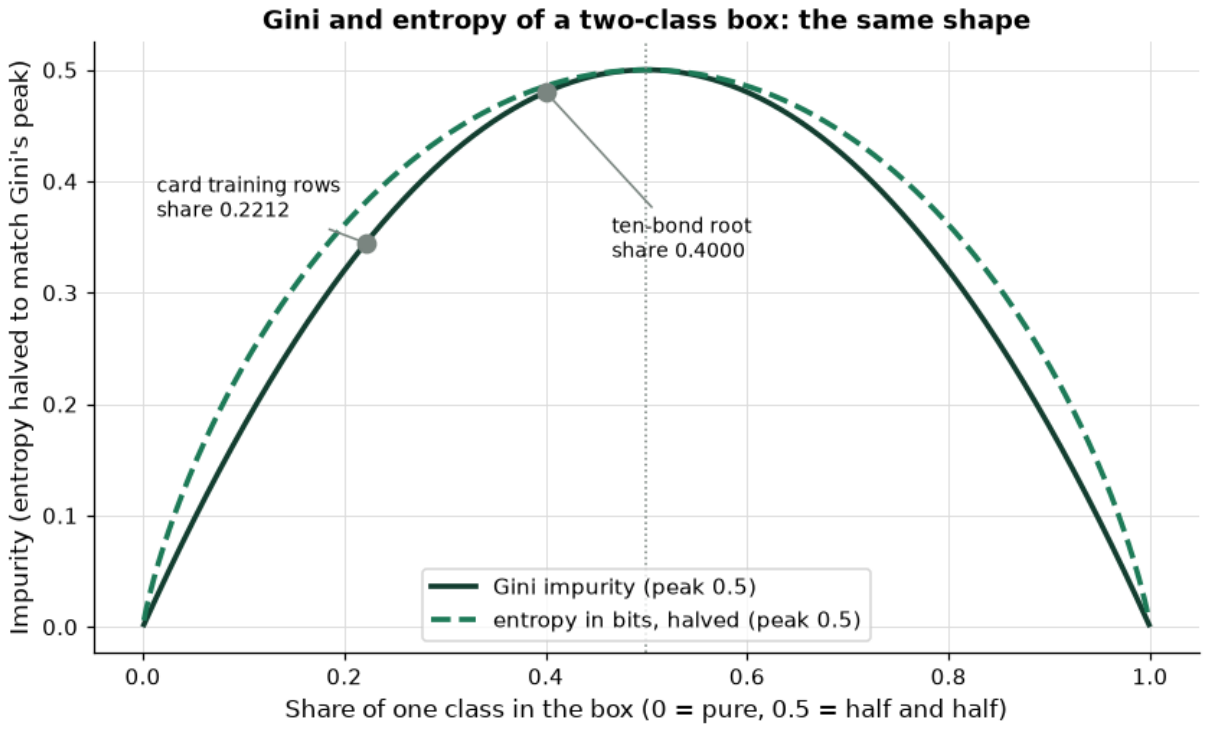
Which question wins, and scikit-learn agrees





Winner: leverage <= 3.75, decrease 0.1633. Best on the other column: coverage <= 7.75, 0.0800. 7 of 7 boxes match scikit-learn's tree.

Entropy and information gain



Entropy, in bits

-(p log2 p + (1 - p) log2 (1 - p)): 0 when pure, 1 bit at fifty fifty

Information gain at the root

0.971 - 0.715 = 0.256 bits: the parent's entropy minus the weighted entropy of the two boxes

The two usually pick the same split

Gini, entropy, and log_loss (the same as entropy) chose the same root question on the card file and on the ten bonds.

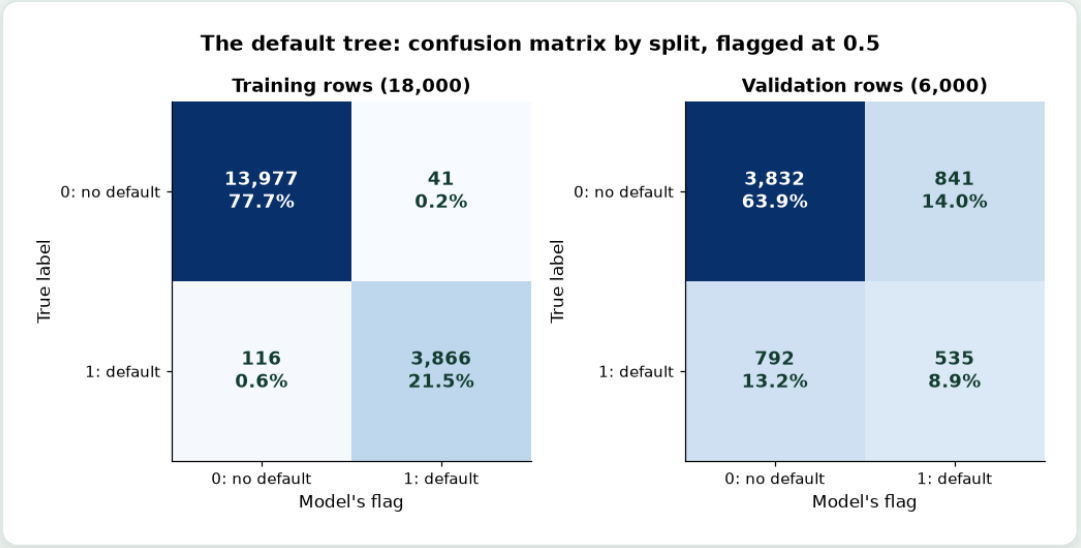
The default tree overfits

38
questions deep

2,750
leaves

6.5
accounts per leaf

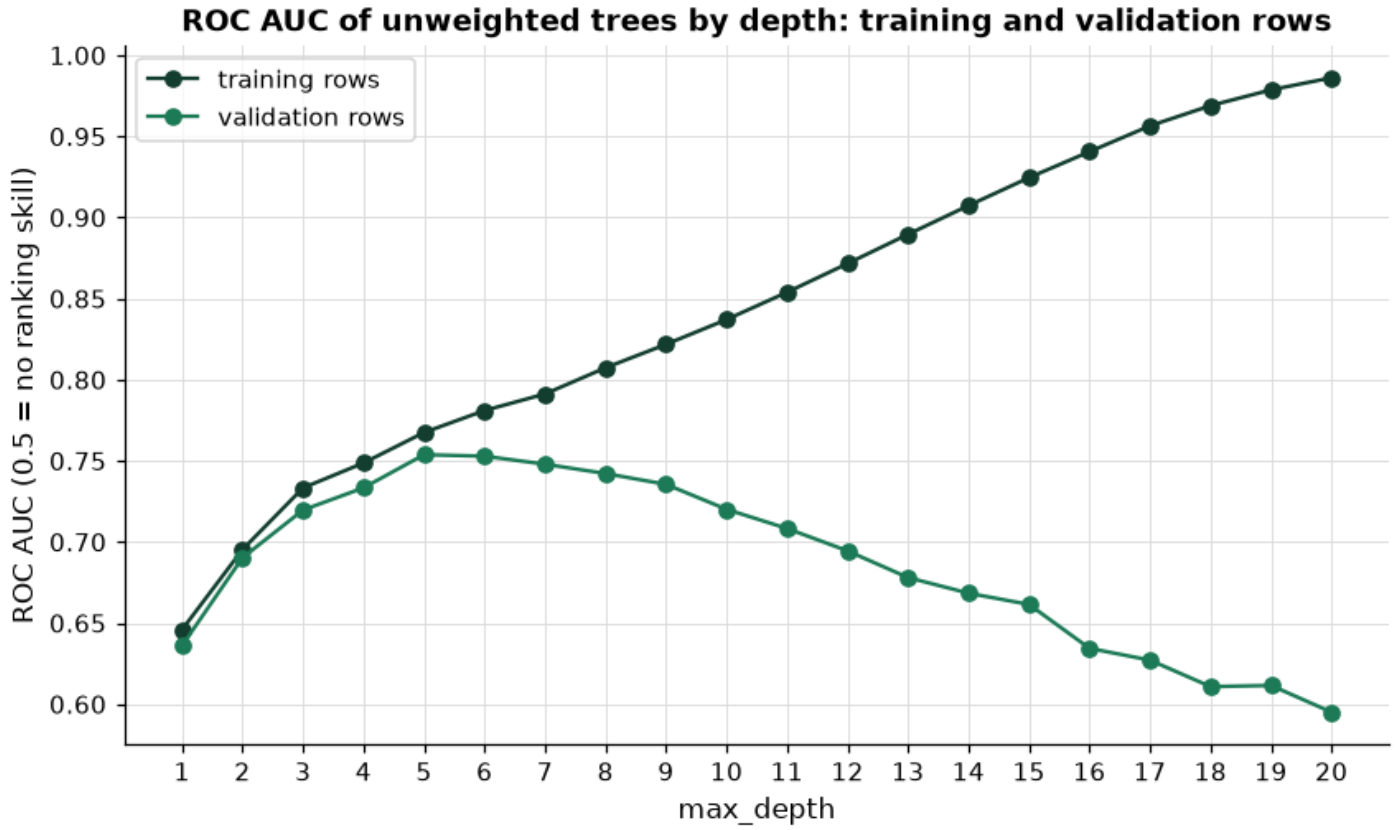
	ROC AUC	Recall	Precision
Training rows	0.9997	0.9709	0.9895
Validation rows	0.6161	0.4032	0.3888



Training recall is not 1: 48 groups of identical accounts (413 in all) hold both outcomes.

Nearly memorised on training rows, much worse on rows it never saw: overfitting.

Overfitting grows with depth



depth 5

validation peak

0.7538

validation ROC AUC there

0.9862

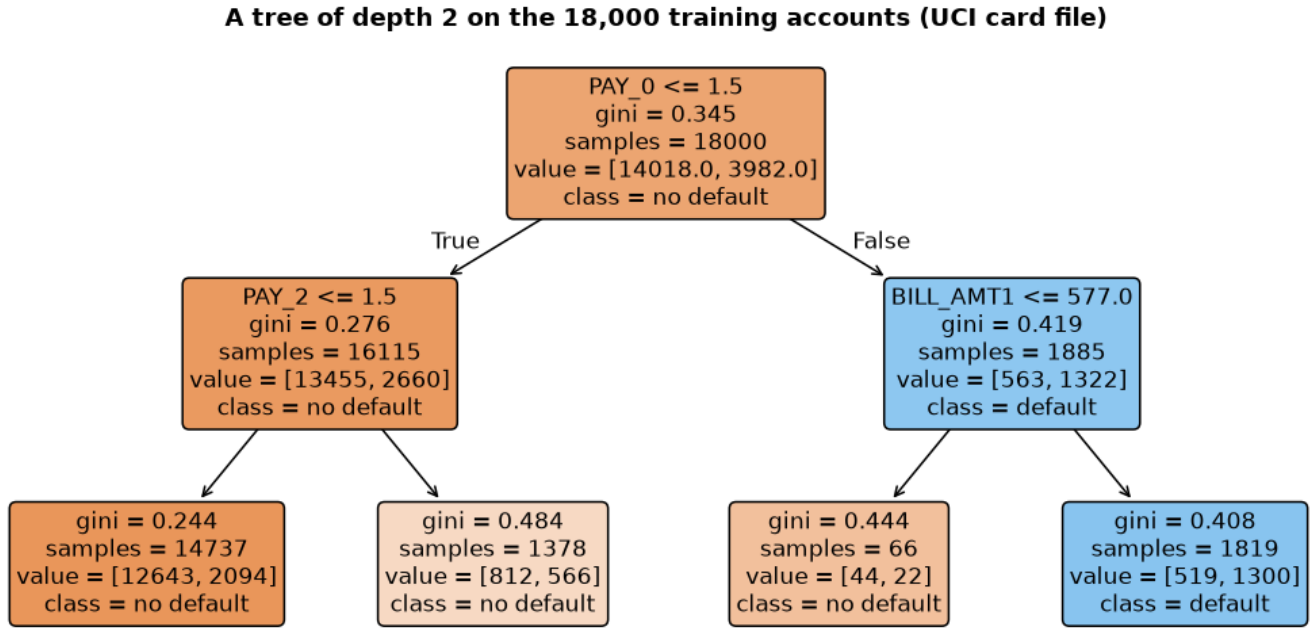
training, depth 20

0.5950

validation, depth 20

The space between the lines is overfitting. The default tree is at depth 38.

Reading a tree: depth 2



In this file's training rows, of accounts with status 2 or more and a bill above 577, 71.5 percent defaulted. A description, not a rule.

Class weights: make the rare class count

2.2602

weight of a default

0.6420

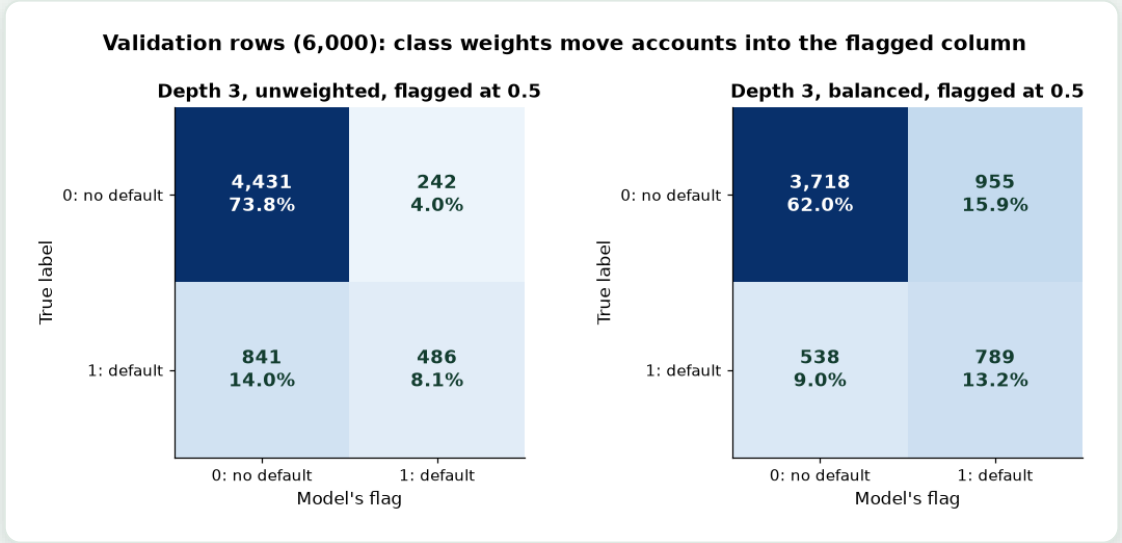
weight of a non-default

9,000

total weight of each class

Depth 3, validation	ROC AUC	Recall	Precision
Unweighted	0.7197	0.3662	0.6676
Balanced	0.7491	0.5946	0.4524

Weighted output is a score: 0.2770 in a leaf of 8,346 accounts where 0.0981 defaulted.

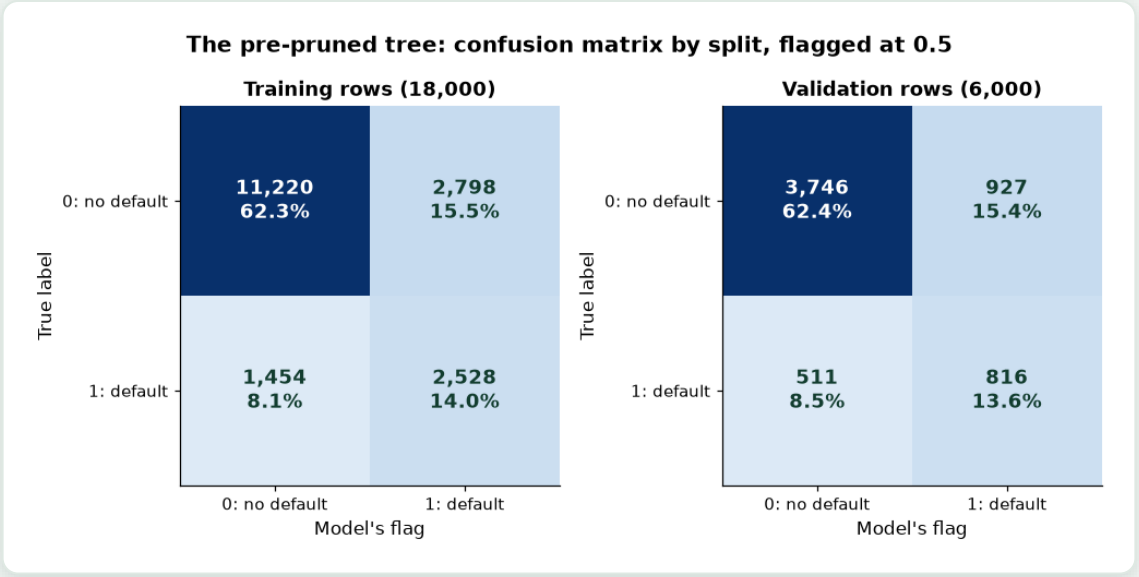


Pre-pruning: stop the tree early

Setting	Stops
`max_depth`	a path longer than this
`min_samples_leaf`	a leaf smaller than this

```
cv=mlkit.cv_folds()   # StratifiedKFold(5,
                      #   shuffle=True, 42)
```

Chosen by five-fold cross-validation inside the 18,000 training rows. 38 tests pass.



depth 8

leaves of at least 200

0.7664

cross-validated ROC AUC

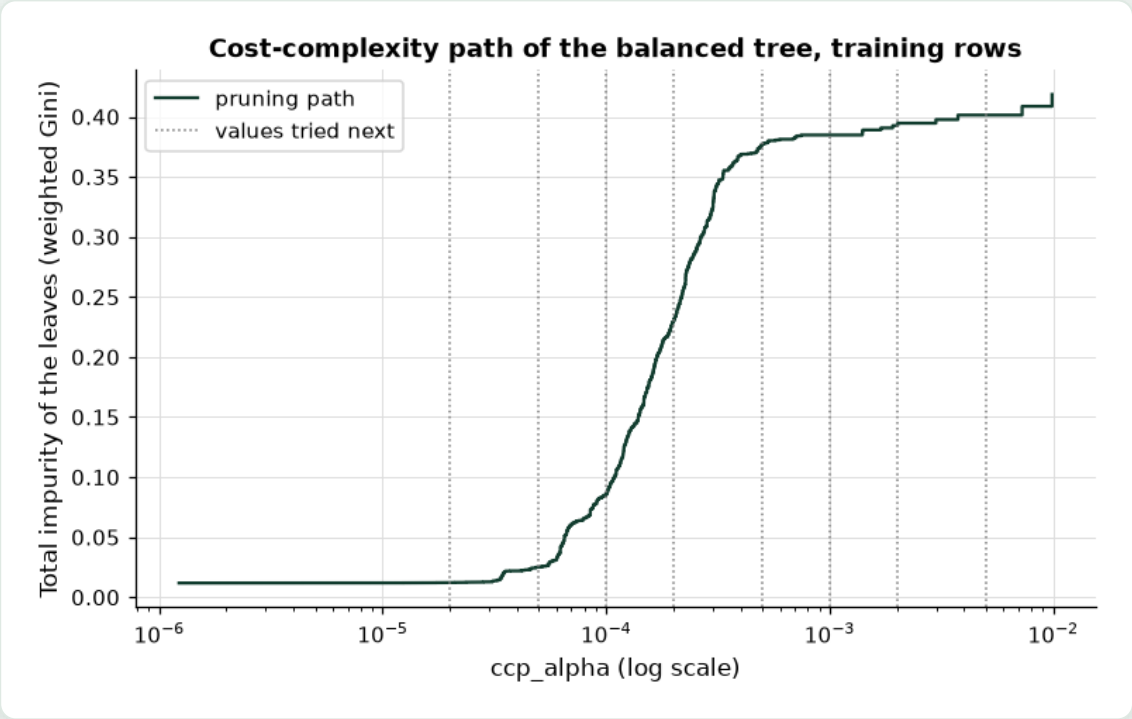
52

leaves

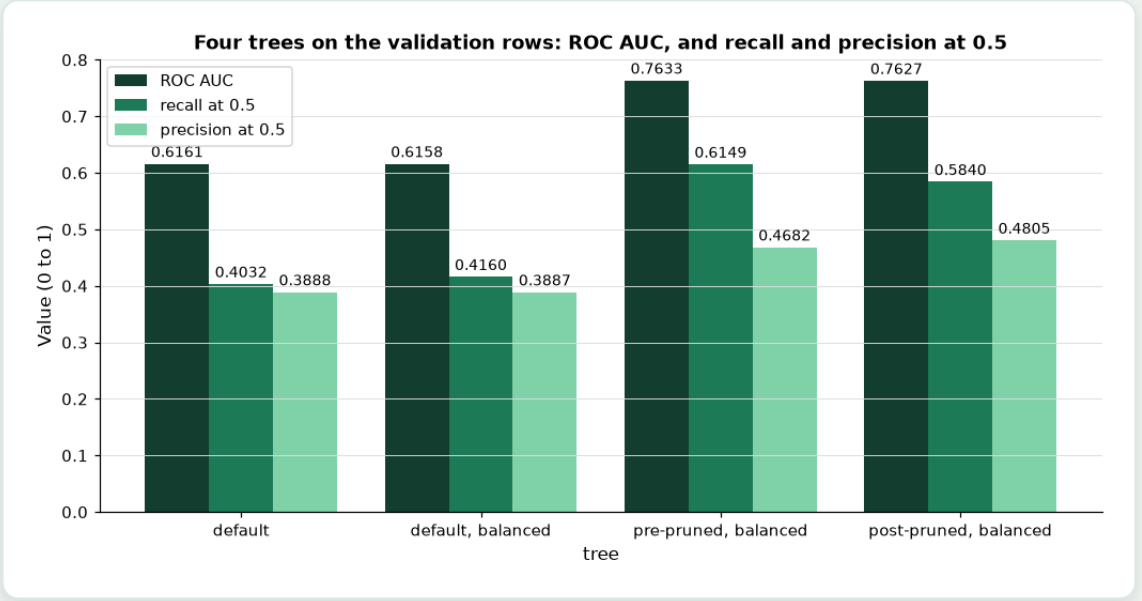
0.7633

validation ROC AUC

Post-pruning: grow, then cut back

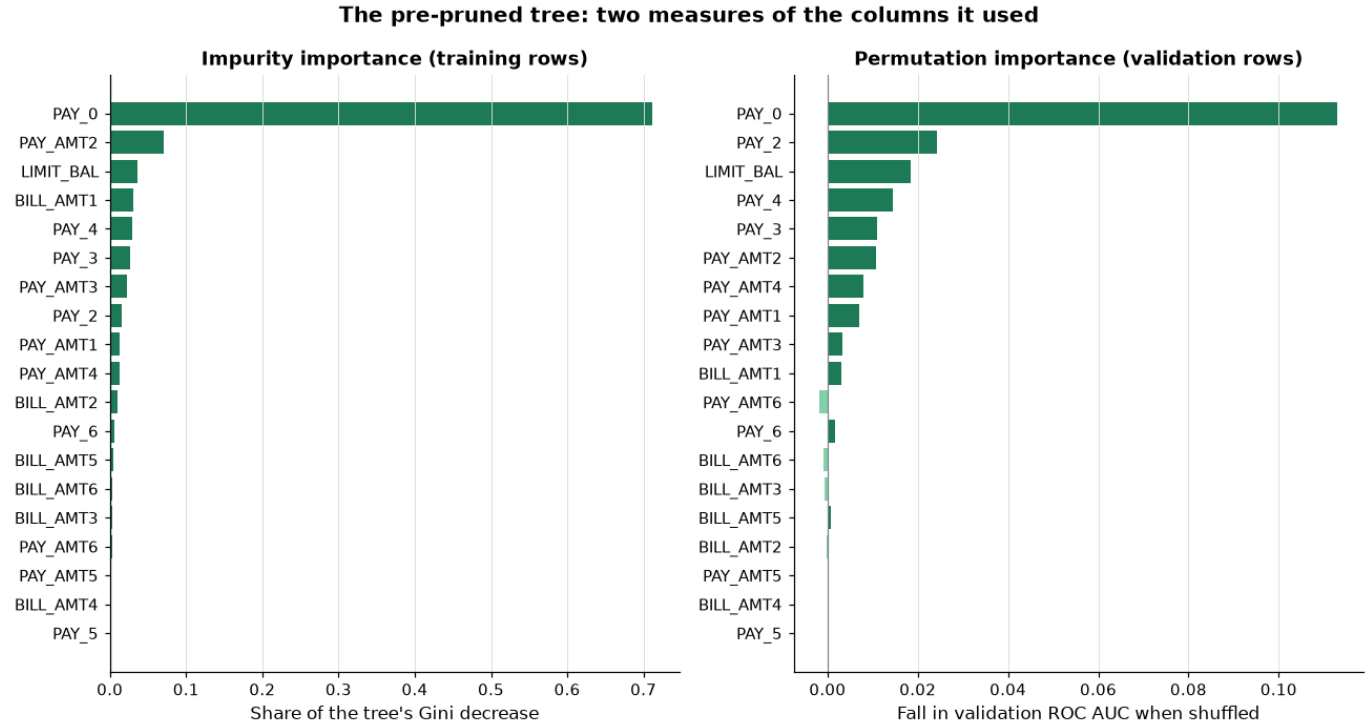


1,743 alphas; best ccp_alpha 0.001 (cross-validated 0.7598). Post-pruned: 12 leaves, validation ROC AUC 0.7627.



Balanced trees ask PAY_0 <= 0.5 first: weights change the questions.

Feature importance, and what it does not mean



Not cause

The tree used it to sort accounts

Not direction

Higher value: more or fewer?

Shared

Related PAY_ columns mask each other

Training rows

Impurity favours many split points

Impurity: training rows. Permutation: fall in validation ROC AUC when a column is shuffled.

Scaling does not change a tree

A split asks above or below; scaling keeps the order of values and thresholds.

6,000

validation accounts, same output (pre-pruned tree)

5

accounts exactly on a threshold (default tree)

Same tree, node for node. A tree needs no scaling.

Every setting of DecisionTreeClassifier

Tree shape and size

max_depth, min_samples_leaf, ccp_alpha: tune first. Also max_leaf_nodes, min_samples_split, min_impurity_decrease, min_weight_fraction_leaf.

How a split is chosen

criterion, splitter, max_features: rarely matter for one tree. max_features returns in Course 5.

Randomness and class weights

random_state: always set. class_weight: a choice about which mistake to avoid, section 9.5.

Constraints

monotonic_cst: not used in the course.

13

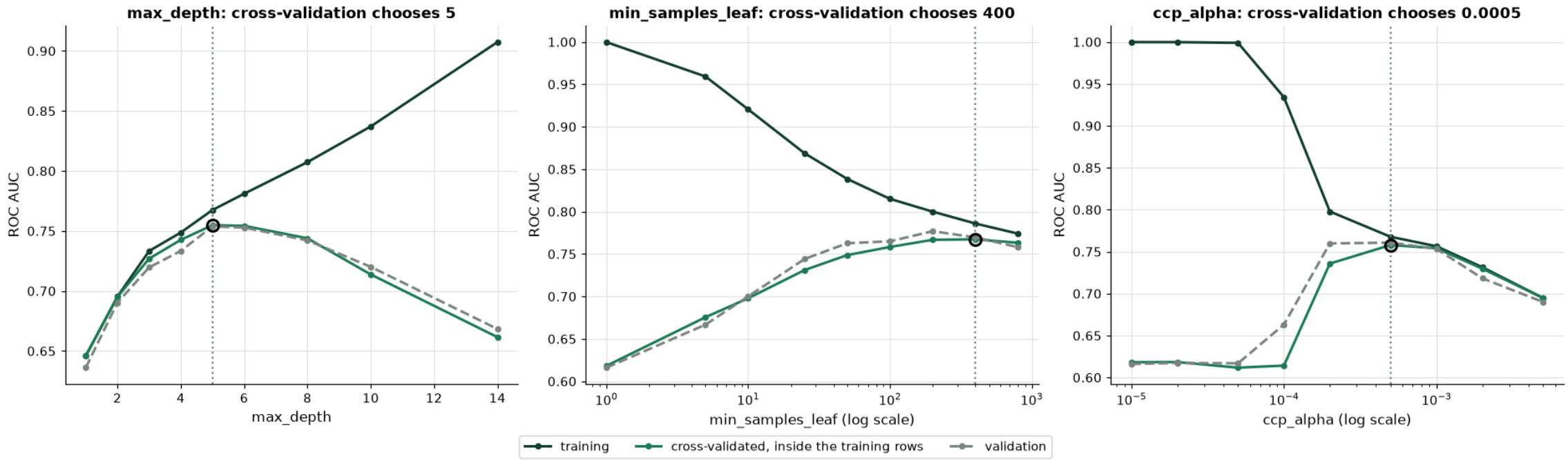
settings in the library

3

to tune first

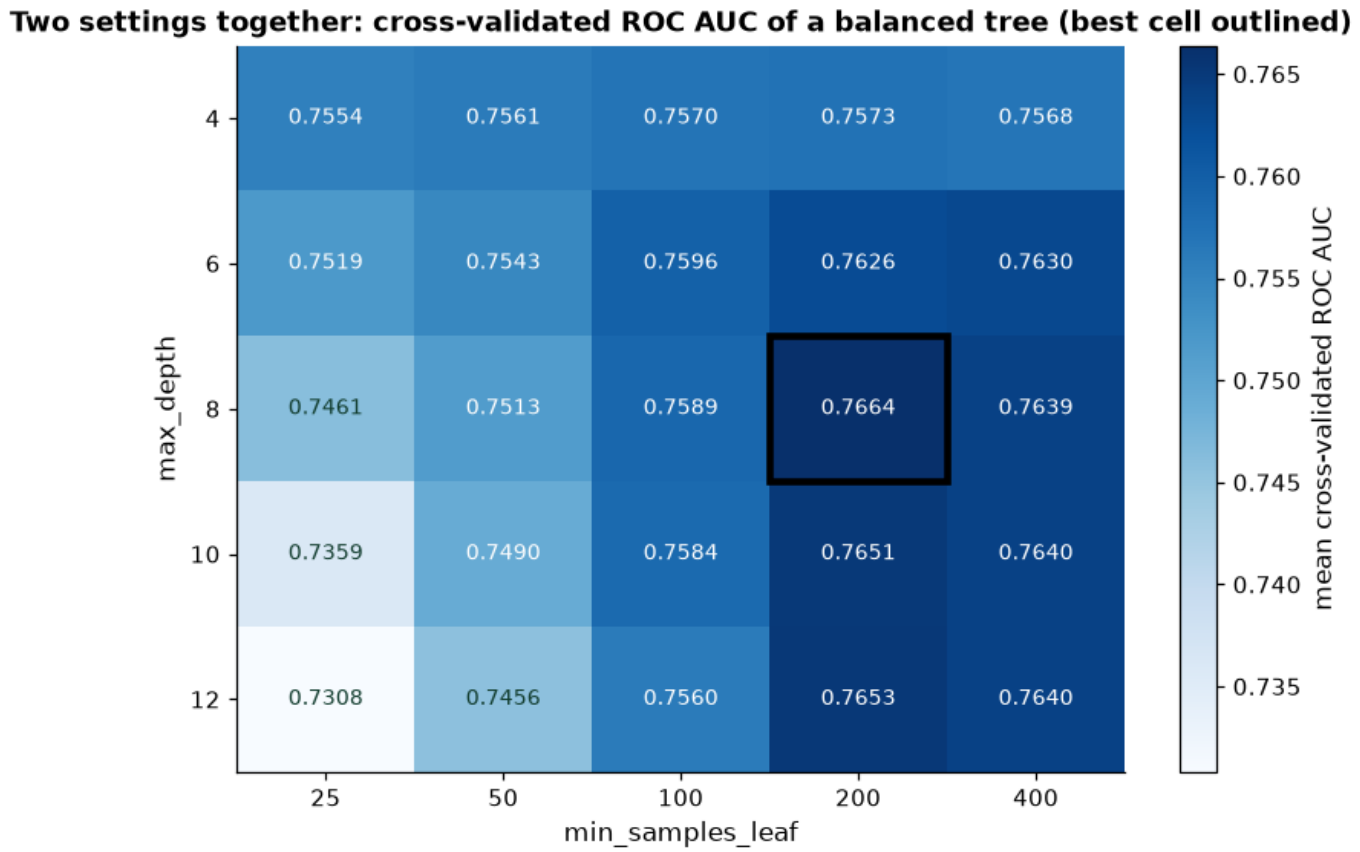
The regressor has the same settings, minus class_weight, and criterion defaults to squared_error.

Sweeping the three settings to tune first



Chosen by cross-validation: max_depth 5, min_samples_leaf 400, ccp_alpha 0.0005. Pick the top of the cross-validated line, never the training line.

Two settings together



8

best depth

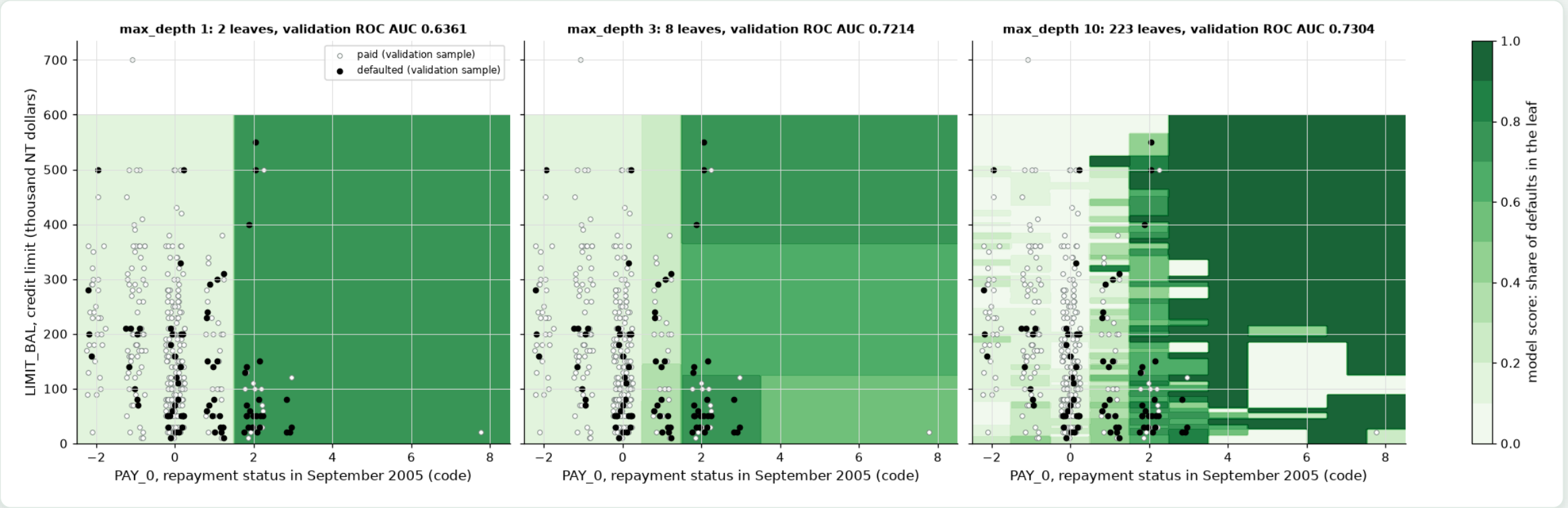
200

smallest leaf

Down a column depth helps, then stops. Along a row bigger leaves help a deep tree most. The worst cells: deep trees with the smallest leaves.

The settings interact, so a search tries them together.

What the settings do to a tree's shape



depth 1: 2 leaves, AUC 0.6361; depth 3: 8 leaves, AUC 0.7214; depth 10: 223 leaves, AUC 0.7304. Every cut is horizontal or vertical.

Excel side by side: a tree by hand

Excel side by side (tested in Excel)

`=COUNTIFS(A2:A18001,"<=1.5")`	16,115 accounts on the left
`=COUNTIFS(A2:A18001,"<=1.5",D2:D18001,1)`	2,660 defaults on the left
`=-(A1/(A1+B1)*LOG(A1/(A1+B1),2)+B1/(A1+B1)*LOG(B1/(A1+B1),2))`	entropy of the 6 / 4 box: 0.9710 bits
`=1-(G2/G1)^2-((G1-G2)/G1)^2`	Gini 0.3446
`=G3/G1*G8+G5/G1*G9  =G7-G10`	0.2906, decrease 0.0539
`=AVERAGEIFS(D2:D18001,A2:A18001,">1.5",C2:C18001,">577")`	leaf share 0.7147
`=IF(A2<=1.5,IF(B2<=1.5,0,0),IF(C2<=577,0,1))`	the tree's flag, all rows
`=H1/(H1+H2)  =H1/(H1+H3)`	recall 0.3052, precision 0.6935

Excel has no decision tree. It can count, weigh, and apply a tree Python fitted.

Save the day: tag the pruned trees

1. Code

mlkit.py and test_mlkit.py

2. Unpruned

results.csv: two trees

3. Pruned

two more rows, then git tag tree-pruned

```
model,rows,roc_auc,average_precision,threshold,accuracy,precision,recall,f1
default,validation,0.6161,0.2917,0.5000,0.7278,0.3888,0.4032,0.3959
"default, balanced",validation,0.6158,0.2926,0.5000,0.7262,0.3887,0.4160,0.4019
"pre-pruned, balanced",validation,0.7633,0.5248,0.5000,0.7603,0.4682,0.6149,0.5316
"post-pruned, balanced",validation,0.7627,0.4954,0.5000,0.7683,0.4805,0.5840,0.5272
```

A tag names one commit for good: git show tree-pruned:results.csv.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the search is fitted on training and validation rows (24,000), then scored on validation: 0.7645.

Code that runs is not the same as code that is right.

What you can do now

- A tree picks the question with the largest Gini decrease
- Left to grow it memorises: 0.9997 training, 0.6161 validation
- Class weights make defaults count; the output becomes a score
- Prune before or after, chosen by cross-validation on training rows
- Importance says what the tree used, never why

Week 2, Day 5 Case study: the card file, a threshold for recall 0.70, test rows opened once